

Telecom Network Performance Analysis – Cell A

Given KPI Values:

KPI	Measured Value	Recommended Value	Status
RSRP	-108 dBm	>-90 dBm	Critical
SINR	8dB	>20 dB	Critical
Latency	90 ms	<20 ms	Critical
Packet Loss	5%	<1%	Critical
Throughput	35 Mbps	>100 Mbps	Critical

1) Network Health Status

The overall health of Cell A is classified as Critical (Congested/Poor Performance)

Reason:

All five KPIs fall within the critical range:

- Very weak signal strength (RSRP)
- High interference (SINR)
- Excessive network delay (Latency)
- Significant packet loss
- Low Throughput

These values indicate that the cell is unable to provide acceptable network performance and requires immediate optimization.

2) Possible Root Causes

Several factors may contribute to the poor performance of Cell A:

Weak Signal Strength

- Long distance from the base station
- Buildings or physical obstacles blocking radio signals
- Poor antenna alignment
- Low transmission power

High Interference

- Neighboring cell interference
- Frequency overlap
- High radio noise
- Improper frequency planning

Network Congestion

- Too many connected users
- Limited radio resources
- Insufficient bandwidth

Backhaul Issues

- Slow or overloaded backhaul connection
- Limited transport capacity between the gNB and the core network

Hardware or Configuration Problems

- Faulty antenna or radio equipment
- Incorrect network configuration

- Software issues in the base station

3) Corrective Actions

The following actions can improve performance of Cell A:

Radio Optimization

- Adjust antenna tilt and orientation
- Increase transmission power where appropriate
- Optimize radio parameters

Reduce Interference

- Reconfigure
- Optimize frequency allocation
- Improve beamforming configuration

Capacity Expansion

- Add additional carriers
- Deploy Small Cells
- Increase available bandwidth
- Upgrade Massive MIMO configuration

Backhaul Improvement

- Upgrade transport links
- Increase backhaul bandwidth
- Reduce network bottlenecks

Continuous Monitoring

- Monitor KPIs in real time
- Perform periodic network optimization
- Detect faults proactively

4) Network Functions Affected

- gNB-Weak radio signal, interference, scheduling inefficiency
- AMF-Mobility management may be affected during handovers
- SMF-Poor session quality due to degraded network conditions
- UPF-Increased latency and packet loss during data forwarding

The gNB is the most significantly affected component because it manages the radio link between the user and the network. Poor radio conditions directly reduce signal quality and throughput.

5) KPI Impact on User Experience

Poor KPI values have a direct impact on subscribers:

Voice Calls

- Call drops
- Poor voice quality
- Audio Interruptions

Video Streaming

- Frequent buffering
- Lower video quality
- Playback interruptions

Online Gaming

- High ping

- Increased lag
- Delayed response

Web Browsing

- Slow page loading
- Delayed downloads
- Poor browsing experience

Video Conferencing

- Frozen video
- Audio delays
- Connection instability